

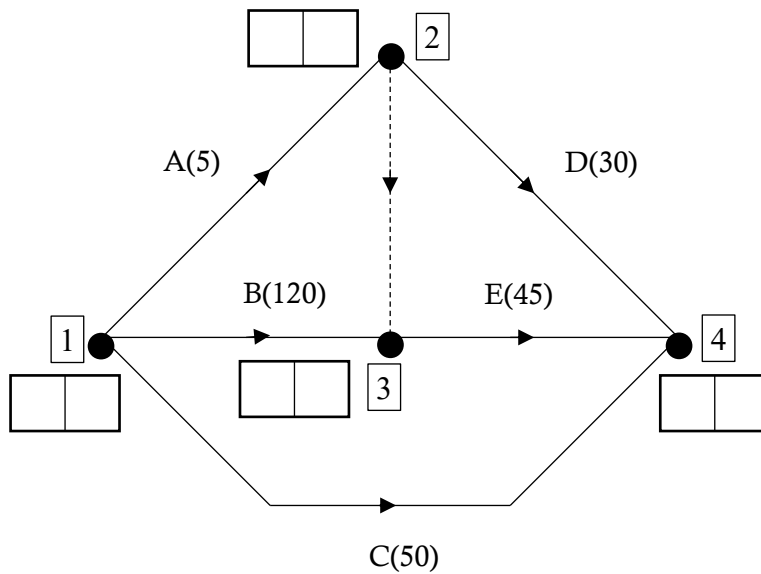
FORWARD AND BACKWARD PASSES

03/02/26

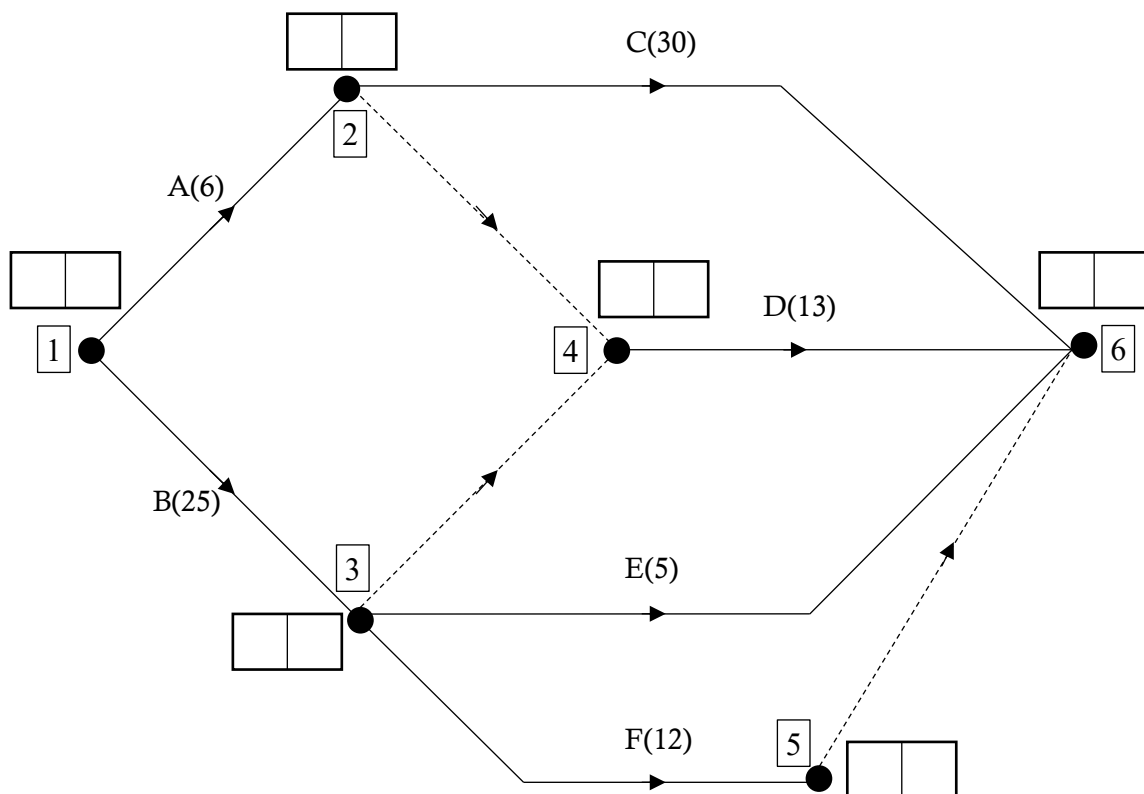
Exercise 2

Perform a forwards and backwards pass on the following activity-on-arc diagrams to find the earliest start time and latest finish time of each activity. Find the minimum total length of time that the project must take, and identify the critical activities.

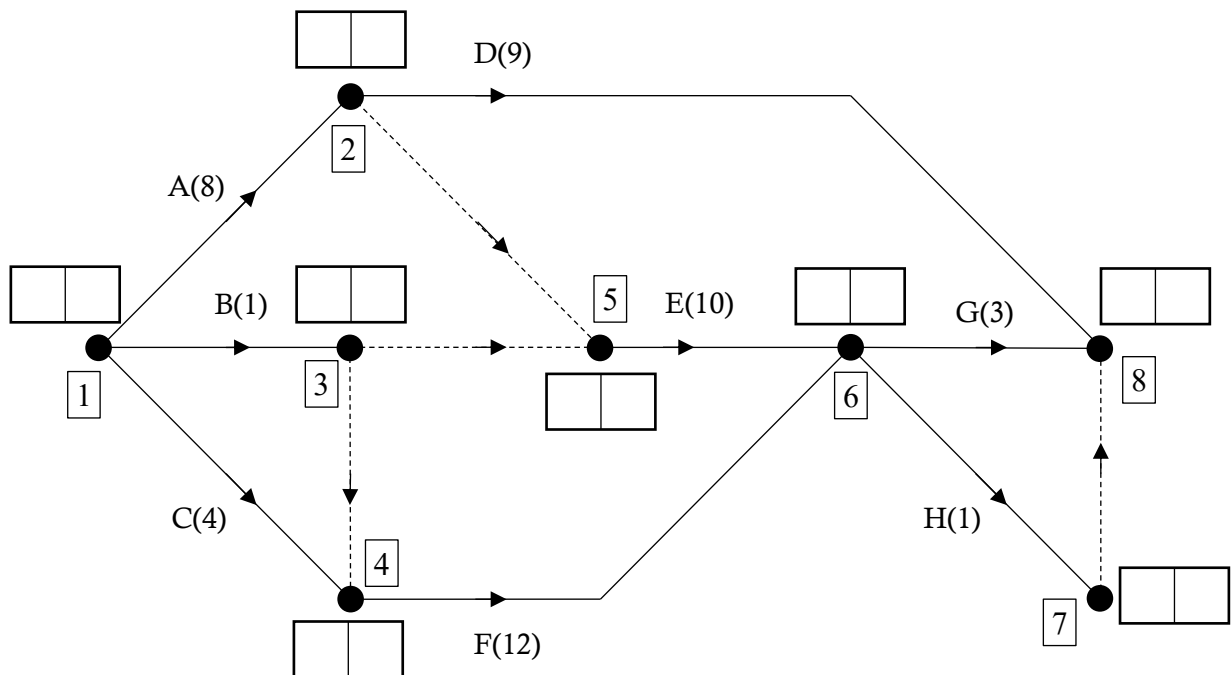
1.



2.

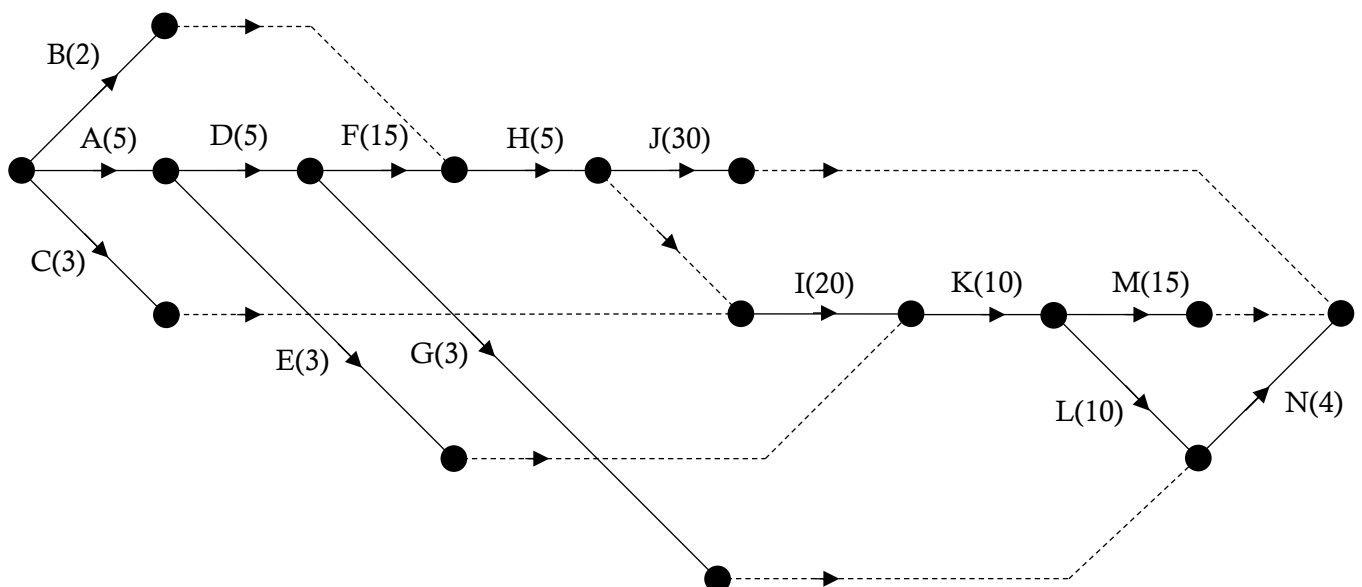


3.



Exercise 3

Someone working on a challenging activity-on-arc problem produces the diagram shown below. Because they put every activity between its own pair of events, they have needed to use a lot of unnecessary dummy activities to indicate precedence.



1. Redraw this network using the minimum possible number of dummy activities. Number the events in an appropriate way.
2. Perform a forwards and backwards pass on your network to find the critical activities and the total time required to complete the project.